MANOJ KUMAR

E- mail: manojsiri880@gmail.com
Mobile: +91-7729012138.

OBJECTIVE:-

Experienced C++ Developer looking for a challenging role to apply skills in application development, project management, and software maintenance, and to grow expertise in C++, Qt, and embedded systems.

PROFESSIONAL PROFILE:-

- Experienced **C++ and Embedded Software Developer** with over **5 years** of expertise in **Infotainment HMI application development**, **Adaptive AUTOSAR**, and **Software Defined Vehicle (SDV)** platforms, Specialized in **system-level programming**.
- Strong background in **automotive embedded systems**, specializing in **Embedded C**, **C++ (11/14/17)**, and **Python**, with extensive experience in **Linux**, **multithreading**, and **RTOS environments**.
- Worked with **Generative AI**.
- Proficient in **AGILE methodologies** and the **Software Development Life Cycle (SDLC)**, ensuring timely and high-quality software delivery.
- Skilled in **bug fixing**, **maintenance**, and **feature enhancement**, with hands-on experience using tools such as **GIT**, **Jenkins**, **JIRA**, **DLT Viewer**, and **Wire shark**.
- Hands-on expertise in **Adaptive AUTOSAR service layer development**, including **time synchronization**, **log and trace**, and **BSW configuration** using **Davinci tools**.
- Adept at developing **automotive infotainment systems**, integrating **HMI modules**, and working with communication protocols like **Bluetooth**, **SPI**, **UART**, **I2C**, and **TCP/IP**.
- Strong analytical, debugging, and problem-solving abilities with deep understanding of **compilation and linking processes**, **STL**, and **software design patterns**.
- Proficient in **Qt-based UI development**, with experience creating efficient, scalable, and user-focused software solutions.
- Recognized as a **proactive and dedicated professional** with a strong sense of **ownership**, **team collaboration**, and consistent performance under pressure.

ACADEMIC CREDENTIALS:-

- **Bachelor of Technology (B. Tech):** From Andhra University of Computer Science Engineering, Visakhapatnam during 2015-2019 affiliated to Andhra University.
- **Intermediate (standard XII):** From Rayalaseema junior college, Srikalahasti during 2013-2015, affiliated to the intermediate Board A.P.
- **Secondary school certificate (SSC):** From Vivekananda E.M School, Srikalahasti, during2013 affiliated to SSC Board A.P.

PROFESSIONAL QUALIFICATION:-

Technology Known: Embedded C, C++11, 14, 17 and Python, Linux.DS.

Other Tools : Davinci Generators, Docker, Wire shark, Git, JIRA, Beyond Compare, Visual Studio, Putty, Notepad++, HMI Artist, Yocto, Make files & CMake, GStreamer, Qt, QML, QNX, I2C, Jenkins, Calculus, RMS, L2B (Label to Binary Patch Tool), ChatGPT, Latex, Anaconda, Tortoise Git, Pintor, PCAT, Alpha, Osiris, Generative AI.

WORK EXPERIENCE:

QUALCOMM

Nov 2023 – Till Date

Role: Sr. System Software Engineer 1

Project Details:

Title :**SDV (Software-Defined Vehicle)**.

Language : C++, QNX, Android

Team Size 52

Description:

A Software-Defined Vehicle (SDV) is a modern vehicle where software drives core functionality, enabling smarter and more flexible vehicle operations. Unlike conventional vehicles that rely mainly on hardware, SDVs use centralized computing systems to control and optimize multiple features, supporting over-the-air updates, advanced connectivity, and autonomous or driver-assistance capabilities. They deliver customizable user experiences, integrate with digital ecosystems, and use data-driven insights to enhance safety, efficiency, and maintenance.

Roles and Responsibilities:

- **Built and maintained software images** for Android and QNX platforms using shell scripts and CMake, proactively resolving **build-time errors**, **linker issues**, and **configuration inconsistencies** to ensure smooth compilation and deployment.
- **Debugged and stabilized video processing tools**, including resolving **Osiris crashes** on QNX through detailed **system-level analysis and debugging**.
- Worked with **Generative AI**.
- Developed and implemented **robust IP-based communication mechanisms** between QNX and Android, ensuring reliable **data transfer, network stability**, and cross-platform interoperability.
- Specialized in **system-level programming.**
- **Designed and integrated HMI applications**, working extensively with **1H1, DMS, and SVC cameras**, ensuring seamless functionality and user interaction.
- Collaborated with **Pintor and CDRS** for **2D and 3D visualization**, enhancing display rendering, frame processing, and image analysis workflows.
- Worked with **target platforms 8775 and 8225** to flash **Meta, QNX, and Android images**, supporting system updates and integration testing.
- **Analyzed, debugged, and resolved display-related issues**, performing root cause analysis and implementing solutions to optimize visual performance.
- Actively participated in **development, integration, and testing cycles**, contributing to **software design improvements**, performance optimization, and **cross-platform system stability**

STELLANTIS **Jan 2022 – Oct 2023**

Role: Sr. Software Engineer

<u>Project Details:</u>

Title **: Adaptive Autosar**

Languages : Embedded C++, python scripting Team Size

Team Size 48

Description:

The AUTOSAR Adaptive Platform provides a scalable and future-ready foundation for automotive ECUs, enabling independent development and deployment of applications across distributed teams. It offers a flexible runtime environment where applications can be added, updated, or modified dynamically, with the ECU serving as a central server. This capability is essential for advanced automotive systems such as ADAS, which process real-time environmental data to enhance driver safety, vehicle performance, and overall system adaptability.

Roles and Responsibilities:

- Reviewed and analyzed **system requirements and technical documents** to contribute to the design, interface development, and architecture maintenance of the Adaptive AUTOSAR platform.
- Developed and maintained **API applications and Conan build scripts**, actively participating in the **service layer and application-level software development**.
- Configured **Basic Software (BSW) modules**, including **Log & Trace, COM, and Time Synchronization**, in alignment with AUTOSAR standards.
- Implemented **Adaptive AUTOSAR features**, such as Time Sync and Log & Trace, using **C++, QEMU, Yocto, Make files, CMake, JSON, ARXML, Qt, QML, Docker, and Davinci Generators**.
- Leveraged **Docker** to enhance application modularity and update mechanisms, supporting **distributed development workflows** and **ADAS functionalities**.
- Developed, configured, and tested **C++ applications** integrated with ARXML and JSON files, following AUTOSAR coding guidelines for maintainable and scalable software.
- Built and deployed **Linux-based ECU images** using Yocto recipes and compiled code in **POSIX and RTOS environments**, ensuring smooth integration and system performance.
- Collaborated with cross-functional teams to integrate applications using **QML, Docker, and Davinci Generators**, supporting both 2D/3D visualization and real-time system operations.
- Conducted **end-to-end development activities**, including coding, integration, debugging, and validation, ensuring software reliability, performance, and adherence to AUTOSAR standards.

KPIT Dec 2019 –Jan 2022

Role: Software Engineer

<u>Project Details:</u>

Title : **HMI Infotainment Application development**

Team Size : 25

Languages : Embedded C++, C

Description:

Automotive infotainment refers to a comprehensive hardware and software platform designed to deliver audio, video, and interactive services within a vehicle. The automotive head unit, often referred to as the infotainment system, serves as a centralized interface, integrating displays, buttons, and controls to manage multiple information and entertainment functions. These functions may include radio broadcasting, audio and video streaming via wired or wireless connections such as HDMI and Ethernet, or media playback from CD, USB, or other external devices, providing a seamless and interactive user experience.

Roles and Responsibilities:

- Developed, implemented, and maintained **HMI and control modules**, including settings functionality, audio/video features, and interactive interfaces.
- Investigated and resolved **system-level and implementation issues**, performing root cause analysis, bug fixes, and code optimization for stable and efficient performance.
- Managed **bug tracking and issue resolution** using JIRA, ensuring timely fixes and maintaining high-quality software standards.
- Designed and executed **unit, integration, and system-level tests**, validating module functionality, performance, and responsiveness across QNX and Linux platforms.
- Built, compiled, and deployed applications using **Yocto, CMake, Makefiles, GStreamer, QNX, I2C, Jenkins, Visual Studio, and Putty**, integrating features across multiple ECUs and hardware targets.
- Applied **C++ and scripting languages** to develop robust service layer and control logic, adhering to coding standards and AUTOSAR guidelines.
- Enhanced **HMI modules** based on evolving requirements, optimizing user interactions, display controls, and multimedia streaming capabilities.
- Leveraged **multithreading, performance profiling, and memory optimization techniques** to improve application efficiency and responsiveness.
- Supported **hardware-software integration**, ensuring seamless communication between ECUs and peripheral devices over protocols like I2C, SPI, CAN, and Ethernet.
- Contributed to **continuous integration and deployment pipelines**, automating builds, performing regression testing, and enabling rapid software updates.
- Collaborated with cross-functional teams to drive **feature development, system integration, and problem-solving**, ensuring delivery of high-quality automotive infotainment solutions.

PERSONAL DETAILS:-

Name	:	D. Manoj Kumar
Date of birth	:	08-10-1997.
Nationality	:	Indian
Languages known	:	English, Telugu, Hindi, Kanada.
Present Address	:	Bangalore

DECLARATION:-

I confirm that the information provided is true and accurate. I would sincerely appreciate the opportunity to join your esteemed organization and am dedicated to delivering my best performance while exceeding expectations.

PLACE: MANOJ KUMAR